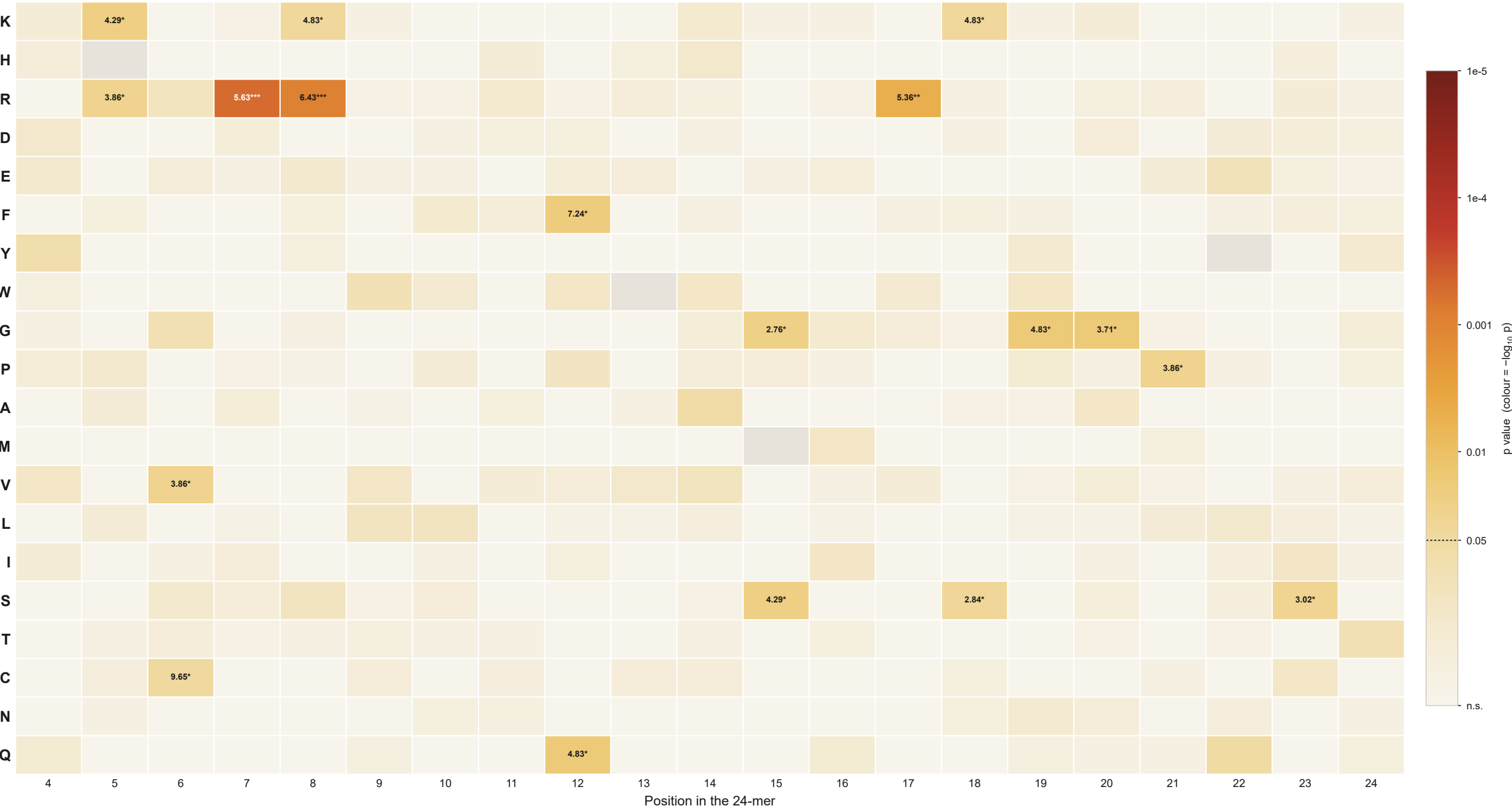


Position × residue enrichment, coloured by significance (Fisher exact)  
colour =  $-\log_{10} p$ , Fisher exact recomputed from the same 2×2 tables — valid at the low cell counts here



n = 20 P/G-D/E/T substrates vs n = 193 non-substrate controls carrying the same motif. Rows are ordered by chemical class — Basic K H R · Acidic D E · Aromatic F Y W · Aliphatic G P A M · Hydrophobic V L I · Polar S T C N Q.

Cell text is the fold change (substrates / controls) of every cell at  $p < 0.05$ , followed by the star bin of its uncorrected p: \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ . Grey = residue absent from both groups, no test possible.

No depleted cell reaches  $p < 0.05$  anywhere in this dataset, so every coloured cell above is an enrichment. 18 of 416 testable cells reach  $p < 0.05$  uncorrected, against 21 expected by chance alone — and none survives FDR correction, so no single cell here is more than a multiple-testing artefact.